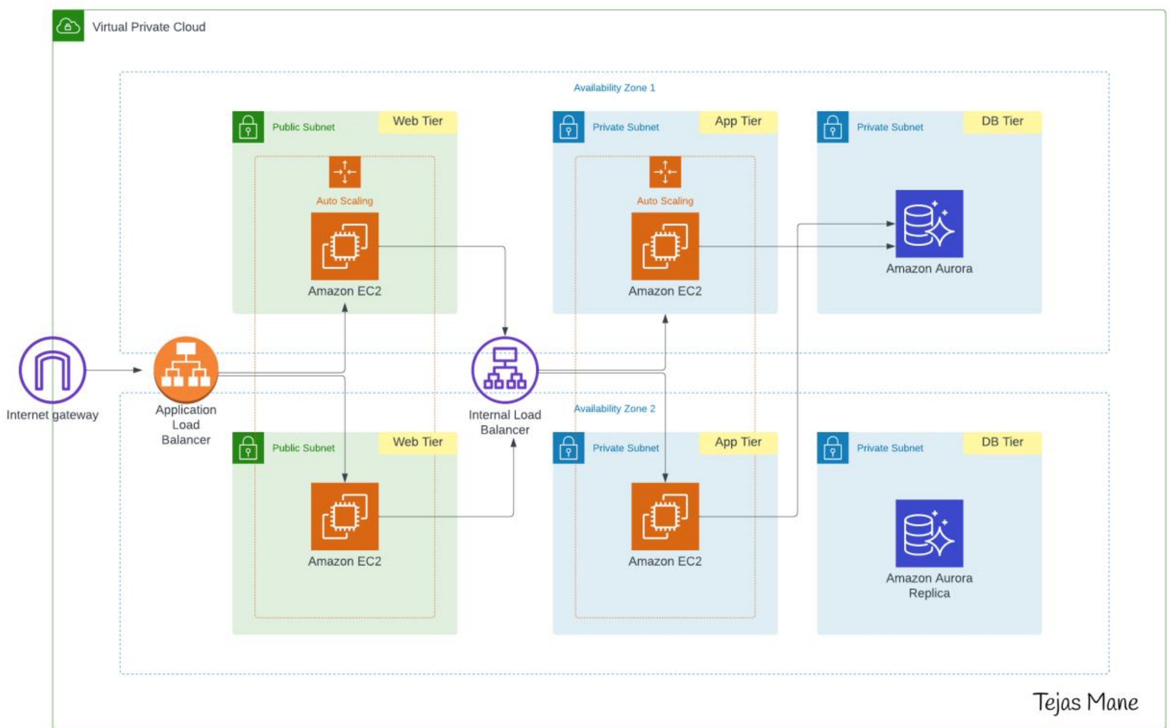
Three-Tier Architecture Overview

A three-tier architecture is a software design pattern that separates an application into three independent layers: the presentation layer, the application layer, and the data layer.

This separation helps improve scalability, security, maintainability, and flexibility, as each layer can be developed, managed, and scaled independently without affecting the others.

Three Tier Web Application



The Three Tiers Explained

1. Presentation (Web) Tier

This is the user-facing layer of the application. It is responsible for displaying content and interacting directly with users through a web browser or client application.

- It handles user input and presents output (UI/UX).
- Technologies typically include HTML, CSS, and JavaScript.

- It communicates with the application tier by sending requests (e.g., API calls).

In an AWS environment, this tier may include:

- An internet-facing Application Load Balancer (ALB)
- Web servers running on EC2 instances or static hosting (e.g., S3 + CloudFront)
- Public subnets with internet access

2. Application (Logic) Tier

This layer contains the core business logic of the application. It acts as the brain of the system.

- Processes requests received from the presentation tier
- Applies business rules and workflows
- Determines how data should be created, updated, or retrieved

In AWS, this tier may include:

- EC2 instances or containers (e.g., ECS/EKS) in private subnets
- An optional internal load balancer for distributing traffic
- No direct inbound internet access (for better security)

This tier may use outbound internet access via NAT if required.

3. Data (Database) Tier

The data tier is responsible for storing and managing application data securely and efficiently.

- Handles data storage, retrieval, and updates
- Ensures data consistency, integrity, and availability

In AWS, this tier typically includes:

- Managed databases such as Amazon RDS or Aurora
- Private database subnets

- Restricted access (only the application tier can communicate with it)

This isolation strengthens security and helps meet compliance requirements.

Request Flow in a Three-Tier Architecture

1. A user accesses the application using a domain (e.g., via DNS like Route 53).
2. The request reaches an internet-facing load balancer.
3. The load balancer forwards the request to the presentation or application tier.
4. The application tier processes the request and applies business logic.
5. If needed, it interacts with the database to fetch or store data.
6. The response is sent back through the layers to the user.

Summary

In simple terms:

- The presentation tier handles what users see
- The application tier handles how the system works
- The data tier handles where data is stored

This layered approach makes applications more secure, scalable, and easier to manage, especially in cloud environments like AWS.